

Multivariate normal
 $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ has density
 $f(\mathbf{x}) = (2\pi)^{-k/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$
Conditional: partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$,
 $\mathbf{X}_1 | \mathbf{X}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{X}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

Conditional moments
Total variance: $\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$.

Random sum: if $N \perp (X_i)$ iid, $\mathbb{E}\left[\sum_{i=1}^N X_i\right] = \mathbb{E}[N] \mathbb{E}[X_1]$.

1 Recap: Poisson Processes & CTMCs

1.1 Poisson process

A **Poisson process** $\{N(t), t \geq 0\}$ with rate $\lambda > 0$ satisfies
(i) $N(0) = 0$; (ii) **independent increments**; (iii) $N(t + s) - N(s) \sim \text{Poisson}(\lambda t)$.

- Interarrivals** $T_i \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ (**memoryless**); arrival times $S_n = \sum_{i=1}^n T_i \sim \text{Gamma}(n, \lambda)$.
- Order-statistic property:** given $N(t) = n, (S_1, \dots, S_n) \stackrel{d}{=} \text{order statistics of } n \text{ iid Uniform}(0, t)$.
- Superposition:** independent rates λ_1, λ_2 merge to rate $\lambda_1 + \lambda_2$.
- Thinning:** label arrivals iid with prob $p \Rightarrow$ Poisson rate λp .
- Competing Poissons:** indep. rates $\lambda_1, \lambda_2, p = \lambda_1/(\lambda_1 + \lambda_2)$,

$$\mathbb{P}(W_n^1 < W_m^2) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} p^k (1-p)^{n+m-1-k}$$

1.2 Non-homogeneous & compound Poisson

A **non-homogeneous Poisson process** (NHPP) with intensity $\lambda(t)$ has independent increments and $\mathbb{P}(N(t+h) - N(t) = 1) = \lambda(t)h + o(h), \mathbb{P}(\geq 2) = o(h)$. Its **mean function** is

$$m(t) = \int_0^t \lambda(s) ds, \quad N(t) \sim \text{Poisson}(m(t)).$$

Thinning by $p(t)$ yields NHPP intensity $\lambda(t)p(t)$. **Time-rescale:** $\tilde{N}(s) = N(\Lambda^{-1}(s))$ is homogeneous PP rate 1, where $\Lambda(t) = \int_0^t \lambda(u) du$.

A **compound Poisson process** $X(t) = \sum_{i=1}^{N(t)} Y_i$ (Y_i iid, indep. of N) satisfies

$\mathbb{E}[X(t)] = \lambda t \mathbb{E}[Y_1], \quad \text{Var}(X(t)) = \lambda t \mathbb{E}[Y_1^2]$.
Merge compound: indep. rates λ_1, λ_2 merge to rate $\lambda_1 + \lambda_2$ with mixture jump law $F = (\lambda_1 F_1 + \lambda_2 F_2)/(\lambda_1 + \lambda_2)$.

Mixed Poisson: $L \sim g, \{X(t) | L = \lambda\} \sim \text{PP}(\lambda)$. Then $\mathbb{E}[N(t)] = t \mathbb{E}[L], \text{Var}(N(t)) = t \mathbb{E}[L] + t^2 \text{Var}(L)$.

1.3 Continuous-time Markov chains

A **continuous-time Markov chain** (CTMC) satisfies the **Markov property** $\mathbb{P}(X(t+s) = j | X(s) = i, \{X(u), u < s\}) = P_{ij}(t)$, depending only on lag t . **Chapman-Kolmogorov:**

$$P_{ij}(t+s) = \sum_k P_{ik}(t)P_{kj}(s), \quad P(t+s) = P(t)P(s).$$

Sojourn time in state i is $\text{Exp}(\nu_i)$; jump to $j \neq i$ with prob q_{ij}/ν_i .

Infinitesimal generator

The **infinitesimal generator** $G = P'(0)$ has
 $G_{ii} = -\nu_i, \quad G_{ij} = q_{ij} \ (i \neq j), \quad \sum_j G_{ij} = 0$.

- Kolmogorov forward:** $P'(t) = P(t)G$.

- Polmogorov backward:** $P'(t) = G^T P(t)$.
- Matrix solution:** $P(t) = e^{Gt} = \sum_{n \geq 0} (Gt)^n / n!$.

1.4 Birth-death & stationary

A **birth-death process** on $\{0, 1, 2, \dots\}$ has $q_{n,n+1} = \lambda_n, q_{n,n-1} = \mu_n, \nu_n = \lambda_n + \mu_n$. **Stationary recurrence:**

$$\frac{\pi_n}{\pi_0} = \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}, \quad \sum_n \pi_n < \infty.$$

M/M/1 queue ($\lambda_n = \lambda, \mu_n = \mu, \rho = \lambda/\mu < 1$):
 $\pi_n = (1 - \rho)\rho^n$.

Stationary & reversibility

π is **stationary** iff $\pi G = 0$, equivalently the **global balance** $\pi_i \nu_i = \sum_{j \neq i} \pi_j q_{ji}$. **Ergodic theorem:** irreducible positive-recurrent $\Rightarrow \lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j$, independent of i .

Detailed balance $\pi_i q_{ij} = \pi_j q_{ji} \Leftrightarrow$ **time reversibility**; implies stationarity.

Embedded chain: if ψ stationary for embedded jump chain (probs q_{ij}/ν_i), then CTMC stationary $\pi_i \propto \psi_i/\nu_i$, equivalently $\psi_i \propto \pi_i \nu_i$.

ODE toolkit (linear systems for CTMC/PP)

Linear 1st-order: $y' = ay + b \Rightarrow y = -b/a + Ce^{at}$ ($a \neq 0$); integrating factor e^{-at} .

2-state CTMC: for $G = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$,

$$P_{00}(t) = \frac{\mu + \lambda e^{-(\lambda+\mu)t}}{\lambda + \mu}, \quad P_{01}(t) = \frac{\lambda(1 - e^{-(\lambda+\mu)t})}{\lambda + \mu}$$

Diagonalisation: $G = UDU^{-1} \Rightarrow P(t) = Ue^{Dt}U^{-1}$; one eigenvalue is 0 with left eigenvector $= \pi$.

Forward equation for PP ($P_n(t) = \mathbb{P}(N(t) = n)$): $P'_n = \lambda P_{n-1} - \lambda P_n, P'_0 = -\lambda P_0$ recovers $P_n(t) = (\lambda t)^n e^{-\lambda t} / n!$.

Yule process (pure birth, rate $n\lambda$ in state n , start at 1): $P_n(t) = e^{-\lambda t} (1 - e^{-\lambda t})^{n-1}$ (geometric on $\{1, 2, \dots\}$).

First-passage mean: $m_i = \mathbb{E}_i[T_A]$ solves $(Gm)_i = -1$ for $i \notin A, m_i = 0$ on A .

PGF method: for birth-death, $G(z, t) = \mathbb{E}[z^{X(t)}]$ reduces $\partial_t G$ to a 1st-order linear PDE in (z, t) ; solve via characteristics.

2 Martingales

2.1 Definitions

A **filtration** is an increasing sequence of σ -fields $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ on Ω . A process $\{X_n\}$ is **adapted** to $\{\mathcal{F}_n\}$ if $\sigma(X_n) \subseteq \mathcal{F}_n$. The **canonical filtration** is $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ – the smallest filtration to which $\{X_n\}$ is adapted.

A process $\{M_n\}$ is a **martingale** w.r.t. $\{\mathcal{F}_n\}$ if:

- Integrable:** $\mathbb{E}[M_n] < \infty$ for all n .
- Adapted:** M_n is $\mathcal{F}_n$ -measurable.
- Fair:** $\mathbb{E}[M_{n+1} | \mathcal{F}_n] = M_n$.

Replace $=$ with $\geq$ for a **submartingale** (drifts up) and $\leq$ for a **supermartingale** (drifts down). **Iterated property:** for $k \geq 1$,

$$\mathbb{E}[M_{n+k} | \mathcal{F}_n] = M_n,$$

so given the past, the expected future increment is 0. A martingale need *not* be a Markov chain; if it is one, $\mathbb{E}[M_{n+1} | M_n] = M_n$ suffices.

Standard examples

- Let ξ_i iid, $S_n = \sum_{i=1}^n \xi_i, S_0 = 0$. The following are martingales w.r.t. $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$:
- Mean-zero sum:** $M_n = S_n$, when $\mathbb{E}[\xi_i] = 0$.

- Compensated square:** $M_n = S_n^2 - n\sigma^2$, when $\text{Var}(\xi_i) = \sigma^2$.
- Compensated cube:** $M_n = S_n^3 - 3nS_n$, for symmetric ± 1 random walk.
- Asymmetric exponential:** $M_n = (q/p)^{S_n}$ for $\xi_i = \pm 1$ with $\mathbb{P}(\xi = 1) = p, q = 1 - p$. This is *the* martingale for asymmetric gambler's ruin.
- Compensated Poisson** (continuous time): $M(t) = N(t) - \lambda t$ for N a PP rate λ .

Note $e^{\lambda S_n}$ alone is **not** a martingale; one must normalise to the **Wald martingale** $M_n = e^{\lambda S_n} / M_\xi(\lambda)^n$, where $M_\xi(\lambda)$ is the MGF of ξ_i .

2.2 Stopping times & OST

A **stopping time** is $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ with $\{T = n\} \in \mathcal{F}_n$, i.e. deciding whether to stop at time n uses only $X_0, \dots, X_n$. Canonical examples: first hitting time $T = \inf\{n : X_n = s\}$ and first exit time $T = \inf\{n : X_n \notin A\}$.

Optional Stopping Theorem: if $\{M_n\}$ is a martingale and T a stopping time with

- $\mathbb{P}(T < \infty) = 1$, and
 - $\{M_{n \wedge T}\}$ is **uniformly integrable**,
- then
- $\mathbb{E}[M_T] = \mathbb{E}[M_0]$
- .
- Sufficient condition**
- for (ii):
-
- $\mathbb{E}[|M_T|] < \infty$
- and
- $\mathbb{E}[|M_n| \mathbf{1}_{T > n}] \rightarrow 0$
- as
- $n \rightarrow \infty$
- .
-
- Boundedness of
- $M_{n \wedge T}$
- (e.g. between absorbing barriers) makes both trivial. Interpretation: no stopping rule turns a fair game into a profit.

Wald's identities

For X_i iid with mean μ , variance $\sigma^2, S_n = \sum_{i=1}^n X_i$, and stopping time T with $\mathbb{E}[T] < \infty$:

$$\text{Wald I : } \mathbb{E}[S_T] = \mu \mathbb{E}[T], \\ \text{Wald II : } \mathbb{E}[(S_T - T\mu)^2] = \sigma^2 \mathbb{E}[T].$$

Wald I is OST applied to $S_n - n\mu$; Wald II to $(S_n - n\mu)^2 - n\sigma^2$.

2.3 Gambler's ruin

Gambler's ruin: $X_0 = a, X_{n+1} = X_n + \xi_{n+1}$ with $\xi = \pm 1, \mathbb{P}(\xi = 1) = p$, absorbing barriers at 0 and N . Let $T = \inf\{n : X_n \in \{0, N\}\}$ and apply OST.

Symmetric ($p = 1/2$): X_n itself is a martingale, so $a = \mathbb{E}[X_T] = N \mathbb{P}(\text{win})$, giving

$$\mathbb{P}(\text{win}) = \frac{a}{N}, \quad \mathbb{P}(\text{ruin}) = 1 - \frac{a}{N}.$$

Via OST on $X_n^2 - n$: $\mathbb{E}[T] = a(N - a)$.

Asymmetric ($p \neq 1/2$): apply OST to $(q/p)^{X_n}$: $(q/p)^a = \mathbb{P}(\text{ruin}) + (q/p)^N \mathbb{P}(\text{win})$, hence

$$\mathbb{P}(\text{ruin}) = \frac{(q/p)^N - (q/p)^a}{(q/p)^N - 1}.$$

2.4 Inequalities & convergence

Doob's maximal inequality (non-negative submartingale $M_n, a > 0$):

$$\mathbb{P}\left(\max_{k \leq n} M_k \geq a\right) \leq \frac{\mathbb{E}[M_n]}{a}.$$

Doob's L^p inequality ($p > 1$): $\|\max_{k \leq n} M_k\|_p \leq \frac{p}{p-1} \|M_n\|_p$.

Martingale convergence theorem: bounded in L^1 (i.e. $\sup_n \mathbb{E}[M_n] < \infty$) $\Rightarrow M_n \rightarrow M_\infty$ a.s. with $\mathbb{E}[M_\infty] < \infty$.

Branching martingale (Galton–Watson with offspring mean m): Z_n/m^n is a martingale (and $\rightarrow W$ a.s.).

Likelihood-ratio martingale: for $X_i \stackrel{\text{iid}}{\sim} f, L_n = \prod_{i=1}^n g(X_i)/f(X_i)$ is a martingale under f .

3 Renewal Theory

3.1 Setup

A **renewal process** has iid non-negative interarrivals $X_i \sim F$ with mean $\mu = \mathbb{E}[X_1]$ and variance σ^2 . Define **epochs** $S_0 = 0, S_n = \sum_{i=1}^n X_i$, and the **counting process**

$N(t) = \sup\{n \geq 0 : S_n \leq t\}$.
Fundamental relation: $\{N(t) \geq k\} \Leftrightarrow \{S_k \leq t\}$, so
 $\mathbb{P}(N(t) = k) = F_k(t) - F_{k+1}(t)$,

where F_k is the k -fold convolution of F . The **renewal function** is

$$m(t) = \mathbb{E}[N(t)] = \sum_{k \geq 1} F_k(t).$$

SLLN for renewals: $N(t)/t \rightarrow 1/\mu$ a.s. as $t \rightarrow \infty$, for $\mu < \infty$.

Poisson special case: $X_i \sim \text{Exp}(\lambda)$ gives $S_k \sim \text{Gamma}(k, \lambda)$ and $m(t) = \lambda t$.

3.2 Renewal equation & main theorems

Renewal equation

Conditioning on X_1 :

$$m(t) = F(t) + \int_0^t m(t-s) dF(s).$$

Equivalent convolution form: $m(t) = F(t) + \int_0^t F(t-y) dm(y)$.

Laplace transform: $\tilde{m}(s) = \tilde{F}(s)/(s(1 - \tilde{F}(s)))$.

Asymptotic theorems

- Elementary renewal theorem (ERT):** $m(t)/t \rightarrow 1/\mu$.
- Asymptotic variance:** $\text{Var}(N(t)) \sim \sigma^2 t / \mu^3$ for $\sigma^2 < \infty$.
- Blackwell's theorem** (F non-arithmetic, $\mu < \infty$): for any $h > 0$,

$$m(t+h) - m(t) \rightarrow h/\mu.$$

Arithmetic span d : $m(nd) \sim nd/\mu$.

- Key renewal theorem (KRT):** if g is **directly Riemann integrable (DRI)** on $[0, \infty)$ and $h(t) = g(t) + \int_0^t h(t-s) dF(s)$, then $h(t) = \int_0^\infty g(t-s) dm(s)$ and

$$h(t) \rightarrow \frac{1}{\mu} \int_0^\infty g(u) du.$$

- CLT:** for $\sigma^2 < \infty$,
 $\frac{N(t) - t/\mu}{\sqrt{\sigma^2 t / \mu^3}} \xrightarrow{d} \mathcal{N}(0, 1)$.

3.3 Renewal-reward

A **renewal-reward** process has iid pairs (X_i, R_i) (within a pair X_i, R_i may be dependent) with cumulative reward

$$R(t) = \sum_{i=1}^{N(t)} R_i.$$

Renewal-reward theorem: under $\mathbb{E}[R_1], \mu < \infty$,
 $\frac{R(t)}{t} \xrightarrow{\text{a.s.}} \frac{\mathbb{E}[R_1]}{\mathbb{E}[X_1]}, \quad \frac{\mathbb{E}[R(t)]}{t} \rightarrow \frac{\mathbb{E}[R_1]}{\mathbb{E}[X_1]}.$

Sketch: $R(t)/t = (R(t)/N(t))(N(t)/t) \rightarrow \mathbb{E}[R_1] \cdot 1/\mu$.

3.4 Delayed & stationary renewal

Delayed renewal: first epoch $Y \sim G$ (arbitrary), $X_2, X_3, \dots$ iid F . Then

$$m_G(t) = G(t) + \int_0^t m_F(t-s) dG(s), \quad m_G(t)/t \rightarrow$$

The asymptotic rate is unaffected by the initial delay.

Equilibrium distribution
The **equilibrium distribution** of F is

$$F_e(x) = \frac{1}{\mu} \int_0^x \bar{F}(y) dy, \quad \text{density } \bar{F}(x)/\mu,$$

where $\bar{F} = 1 - F$.

Stationary renewal: take $Y \sim F_e$. Then $\{N(t+s) - N(s)\}_{t \geq 0}$ has the same law for every $s \geq 0$, and the renewal function is *exact* (not asymptotic):

$$m_s(t) = \frac{t}{\mu} + \frac{\mathbb{E}[X^2]}{2\mu^2} - 1.$$

3.5 Age, excess, spread

For an inspection at time t :

- Age** $A(t) = t - S_{N(t)}$ – time since last renewal.
- Excess** (residual life) $Y(t) = S_{N(t)+1} - t$ – time until next renewal.
- Spread** $L(t) = X_{N(t)+1} = A(t) + Y(t)$ – length of the cycle covering t .

Limiting distributions (F non-arithmetic)

Both age and excess converge in distribution to the equilibrium distribution:

$$A(t), Y(t) \xrightarrow{d} F_e, \quad \text{density } \bar{F}(x)/\mu.$$

Inspection paradox

$$\mathbb{E}[L(t)] \rightarrow \frac{\mathbb{E}[X^2]}{\mu} = \mu + \frac{\sigma^2}{\mu} \geq \mu,$$

with equality iff X_1 is degenerate. Intuitively, a random inspection time is **length-biased** – more likely to fall inside a long cycle than a short one.

3.6 Alternating renewal & regenerative

Alternating renewal: alternating ON/OFF cycles with means μ_1, μ_2 . Long-run ON-fraction

$$\lim_{t \rightarrow \infty} \mathbb{P}(\text{ON at } t) = \frac{\mu_1}{\mu_1 + \mu_2}.$$

Regenerative process: $\{X(t)\}$ regenerates at epoch X_1 (cycle length); for measurable f ,

$$\frac{1}{t} \int_0^t f(X(s)) ds \xrightarrow{a.s.} \frac{\mathbb{E}\left[\int_0^{X_1} f(X(s)) ds\right]}{\mu}.$$

Semi-Markov (gloss): holding-time distribution F_{ij} depends on (current i , next j) state; embedded jump chain governs transitions.

4 Brownian Motion

4.1 Definition & finite-dim

A **Brownian motion** (BM) $\{B(t), t \geq 0\}$ satisfies:

- $B(0) = 0$.
- Stationary, independent increments:** for $0 \leq t_0 < \dots < t_n$, the increments $B(t_i) - B(t_{i-1})$ are independent, and $B(t) - B(s) \stackrel{d}{=} B(t-s)$.
- For $t > 0$, $B(t) \sim \mathcal{N}(0, \sigma^2 t)$.

Standard BM: $\sigma = 1$, so $B(t) \sim \mathcal{N}(0, t)$.

Finite-dimensional distribution

For $0 < t_1 < \dots < t_n$, $(B(t_1), \dots, B(t_n)) \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $\Sigma_{ij} = \sigma^2 \min(t_i, t_j)$.

Covariance kernel: $\text{Cov}(B(s), B(t)) = \sigma^2 \min(s, t)$.

Properties

- Self-similarity:** $\frac{1}{\sqrt{c}} B(ct) \stackrel{d}{=} B(t)$ for $c > 0$.

- Markov property:** for $s < t$, $B(t) \mid B(s) \sim \mathcal{N}(B(s), \sigma^2(t-s))$.
- Reverse conditioning:** for $s < t$, $B(s) \mid B(t) = c \sim \mathcal{N}\left(\frac{cs}{t}, \frac{\sigma^2 s(t-s)}{t}\right)$.
- Linear combinations are Gaussian: $\mathbf{a}^\top (B(t_1), \dots, B(t_n)) \sim \mathcal{N}(0, \mathbf{a}^\top \Sigma \mathbf{a})$.

4.2 Reflection principle & hitting times

The **hitting time** of level a is

$$T_a = \inf\{t > 0 : B(t) = a\},$$

and the running maximum is $M(t) = \max_{s \leq t} B(s)$. **Reflection on principle:** paths reaching a then continuing are in bijection (via reflection about $y = a$ after T_a) with paths ending above a . For **standard** BM and $a > 0$:

$$\mathbb{P}(M(t) \geq a) = 2\mathbb{P}(B(t) \geq a) = 2\left(1 - \Phi(a/\sqrt{t})\right).$$

Joint event: for $b \leq a$ (with $a > 0$),

$$\mathbb{P}(M(t) \geq a, B(t) \leq b) = \mathbb{P}(B(t) \geq 2a - b).$$

Hitting time density (standard BM):

$$f_{T_a}(t) = \frac{|a|}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right), \quad t > 0.$$

The level is hit a.s. but the wait has infinite mean:

$$\mathbb{P}(T_a < \infty) = 1, \quad \mathbb{E}[T_a] = \infty.$$

4.3 BM with drift, geometric BM

Brownian motion with drift, drift μ , volatility $\sigma > 0$:

$$X(t) = \mu t + \sigma B(t), \quad X(t) \sim \mathcal{N}(\mu t, \sigma^2 t).$$

Increments remain Gaussian and independent:

$X(t) - X(s) \sim \mathcal{N}(\mu(t-s), \sigma^2(t-s))$, $s < t$.
Conditional law: $X(t) \mid X(s) \sim \mathcal{N}(X(s) + \mu(t-s), \sigma^2(t-s))$.

Reverse: for $s < t$, $X(s) \mid X(t) \sim \mathcal{N}\left(\mu s + \frac{s}{t}(X(t) - \mu t), \frac{\sigma^2 s(t-s)}{t}\right)$.

Geometric BM

Geometric Brownian motion models positive processes (e.g. asset prices):

$$S(t) = S(0) \exp(\mu t + \sigma B(t)).$$

- Log-normal:** $\log S(t) \sim \mathcal{N}(\log S(0) + \mu t, \sigma^2 t)$.
- GBM mean:** $\mathbb{E}[S(t)] = S(0) \exp\left((\mu + \frac{\sigma^2}{2})t\right)$.
- GBM variance:** $\text{Var}(S(t)) = S(0)^2 e^{2\mu t + \sigma^2 t} (e^{\sigma^2 t} - 1)$.

4.4 Conditional BM

For $s < t$ and standard BM, conditioning on the endpoint gives

$$B(s) \mid B(t) = c \sim \mathcal{N}\left(\frac{cs}{t}, \frac{s(t-s)}{t}\right).$$

Bridge connection: setting $c = 0$ recovers the marginal of the **Brownian bridge** on $[0, t]$ (variance $s(t-s)/t$; on $[0, 1]$ this is $s(1-s)$). On $[0, 1]$ with $B(1) = c$ given, $B(s) \mid B(1) = c \sim \mathcal{N}(cs, s(1-s))$.

4.5 Quadratic variation & Itô (lite)

Quadratic variation: for partitions Π of $[0, t]$ with mesh $\|\Pi\| \rightarrow 0$,

$$\sum_i (B(t_{i+1}) - B(t_i))^2 \xrightarrow{P} t, \quad [B, B]_t = t.$$

Itô's formula (lite, $f \in C^2$):

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt.$$

Two-sided exit (BM with drift): for $X_t = \mu t + \sigma B_t$, $X_0 \in (-a, b)$, apply OST to the exponential martingale at $\theta =$

$-2\mu/\sigma^2$ to get ruin/escape probabilities (mirrors gambler's ruin).
Arcsine law: fraction of $[0, 1]$ that standard BM is positive has density $\frac{1}{\pi \sqrt{x(1-x)}}$, $x \in (0, 1)$.

Donsker (functional CLT): rescaled random walk $S_{\lfloor nt \rfloor} / \sqrt{n} \Rightarrow B(t)$ in $C[0, 1]$.

5 Gaussian Processes

A **Gaussian process** is a stochastic process $\{X(t), t \geq 0\}$ such that for every finite $t_1, \dots, t_n$, the vector $(X(t_1), \dots, X(t_n))$ is multivariate Gaussian. It is fully characterised by the **mean function** $m(t) = \mathbb{E}[X(t)]$ and the **covariance kernel** $k(s, t) = \text{Cov}(X(s), X(t))$.

5.1 Brownian bridge

The **Brownian bridge** on $[0, 1]$ is $X(t) = B(t) - tB(1)$ for standard BM B (equivalently the law of B conditioned on $B(1) = 0$). **Mean and covariance** (for $s \leq t$):

$$\mathbb{E}[X(t)] = 0, \quad \text{Cov}(X(s), X(t)) = s(1-t).$$

Equivalently, $\text{Cov}(X(s), X(t)) = \min(s, t) - st$.

Marginal: at fixed $t \in (0, 1)$,

$$X(t) \sim \mathcal{N}(0, t(1-t)).$$

Joint: for $s \leq t$,

$$(X(s), X(t)) \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} s(1-s) & s(1-t) \\ s(1-t) & t(1-t) \end{pmatrix}\right).$$

Endpoints (pinning): $X(0) = X(1) = 0$ a.s.

Increments NOT independent (the constraint $X(1) = 0$ couples them). The marginal increment law for $s < t < 1$ is

$$X(t) - X(s) \sim \mathcal{N}(0, (t-s) - (t-s)^2).$$

5.2 Integrated Brownian motion

For standard BM $B(t)$, define

$$\text{integrated Brownian motion} \quad Y(t) = \int_0^t B(s) ds.$$

$Y(t)$ is a Gaussian process (intuition: limit of Gaussian Riemann sums). **Mean and covariance** (for $s \leq t$):

$$\mathbb{E}[Y(t)] = 0, \quad \text{Cov}(Y(s), Y(t)) = \frac{s^2(3t-s)}{6}.$$

Variance: $\text{Var}(Y(t)) = t^3/3$, so $Y(t) \sim \mathcal{N}(0, t^3/3)$.

5.3 Ornstein–Uhlenbeck process

The **Ornstein–Uhlenbeck process** is the mean-reverting Gaussian Markov process

$$dX_t = -\theta(X_t - \mu) dt + \sigma dB_t, \quad \theta, \sigma > 0.$$

Solution: $X_t = \mu + (x_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dB_s$.

Mean and variance:

$$\mathbb{E}[X_t] = \mu + (x_0 - \mu)e^{-\theta t}, \quad \text{Var}(X_t) = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}).$$

Stationary distribution: $\mathcal{N}(\mu, \sigma^2/(2\theta))$.

Stationary covariance: $\text{Cov}(X_s, X_t) = \frac{\sigma^2}{2\theta} e^{-\theta|s-t|}$ (depends on $|s-t|$ only).

5.4 Bridge & integrated-BM extras

Time-reversal (standard bridge): $X(1-t) \stackrel{d}{=} X(t)$.

Itô construction: $X(t) = (1-t) \int_0^t \frac{dB(s)}{1-s}$ (standard bridge on $[0, 1]$).

Joint $(B(t), Y(t))$ for $Y(t) = \int_0^t B(s) ds$ (standard BM):

$$\begin{pmatrix} B(t) \\ Y(t) \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} t & t^2/2 \\ t^2/2 & t^3/3 \end{pmatrix}\right).$$

White-noise heuristic (Itô isometry, lite): $\dot{B}$ as “derivative” of BM; for deterministic f ,

$$\int_0^t f(s) dB(s) \sim \mathcal{N}\left(0, \int_0^t f(s)^2 ds\right).$$

Karhunen–Loève (BM on $[0, 1]$): $B(t) = \sum_{n \geq 1} \xi_n \phi_n(t)$, with $\xi_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\phi_n(t) = \sqrt{2} \sin((n - \frac{1}{2})\pi t)/((n - \frac{1}{2})\pi)$.